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RESEARCH PAPER

Comparison of visual acuity charts identifying visual impairment among older people outside the eye clinic

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***Purpose*: Evaluate the construct validity and describe sensitivity, specificity and predictive value of two short charts of visual acuity (VA) and examine whether these can identify and detect signs of visual impairment among older people. *Method*: The study included 43 persons, >65 years, with age related eye disease, living in their own homes. An ophthalmologist assessed the individuals' VA at an eye clinic with the 5 m KM chart. A research assistant assessed individuals' VA by the 1 m KM chart and the Visual Acuity Screening Test in their home environment. *Results*: All persons with a VA level of <0.5 were correctly identified by both instruments. The instruments have good positive and negative predictive values for the 1 m KM chart (73% and 100%) and for the Visual Acuity Screening Test (69% and 100%). The construct validity between the instruments was good, but the assessment at the eye clinic assessed the participants as having higher VA level. *Conclusions*: Both instruments have good construct validity, considering they were carried out in poorer lighting conditions and a good predictive value for screening out VA levels <0.5. The 1 m KM chart showed the best agreement with the 5 m KM chart.**

Keywords: Elderly, visual impairment, vision screening

Implications for Rehabilitation

- Short charts of visual acuity (VA) can be used outside the eye clinic to identify persons with VA level 0.5, in need of rehabilitation, as they are at risk of becoming dependent in daily activities.
- Short charts of VA are portable instruments and useful in other environments than the eye clinic, but training is needed before use.
- Short charts of VA are useful instruments that can be included in Comprehensive Geriatric Assessments (CGA) to identify persons with visual impairment.

Introduction

Trends for the future show a significant growth in the older population [1]. The prevalence of elderly frail people at risk for functional decline, co-morbidity and disability [2,3] is highest among the oldest age groups, which are now the fastest growing [3]. It is not unusual for an elderly person, in combination with morbidity and functional decline, to also have visual impairment. Among community-dwelling adults, the prevalence of visual impairment was found to grow substantially with age, especially among the oldest people [4–6]. In a Finnish population study, 41% of persons aged 75 years or more had a visual impairment [6]. In a considerable number of studies, visual impairment has been associated with different aspects of functional decline, morbidity and disability, such as increased risk of falling [7], fractures [8], depression [9], poorer health [10,11], increased loneliness [11], decreased social network, dependence in daily activities [12] and use of assistive devices [13].

Frailty has become an established concept in research in recent years [3]. Frailty summarizes a state of decreased reserve resistance to stressors as a result of a cumulative decline across multiple physiological systems, hence causing vulnerability strongly related to disability and co-morbidity [3]. Although studies have shown [14–16] that visual impairment is a marker of frailty in the aging population, it is seldom examined in research or in the care or rehabilitation of the elderly population [17,18]. Comprehensive geriatric assessment (CGA) is often seen as a key to evaluating the

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medical, physiological and functional status of frail elderly people, but visual impairment is seldom included as a measure. However, it has been suggested that visual impairment should be assessed for elderly people at risk of falls and hip fracture [19] as well as in a clinical assessment of older people receiving rehabilitation [20]. It has also been stressed that it should be examined routinely in research to shed light on the pathway of disability [20]. To detect the presence of visual impairment, visual acuity (VA) measurement is the one most used, even though it only measures one part of visual functioning [21,22]. Population surveys have shown that a decreased VA level is strongly associated with limitation in performing daily activities [23] and it is thus imperative to find an instrument that can be used outside the eye clinic to identify elderly frail people who have visual impairment in terms of decreased VA.

According to WHO's definition [24,25], people are "visually impaired" when they have a visual acuity level of less than 0.3 or a visual field below 10°. This VA level is often used in healthcare for judging the need of rehabilitation. More recent studies [8,23,26] show that elderly people already have a higher risk of morbidity or mortality at a visual acuity level of 0.5. They are also at risk of becoming more dependent in activities of daily living (ADL) and of using more assistive devices than elderly people with normal vision. Therefore, assessing those who are at risk for frailty in the aging population requires a cut-off level of 0.5 in the screening of frail elderly people. This means that there are both in rehabilitation and research in geriatric settings needs for an instrument that can identify visual impairment as a marker of frailty among elderly people in order to capture those with early signs of visual impairment.

To our knowledge, no study has evaluated an instrument for frail elderly people with the purpose of screening those who show signs of visual impairment, as this is a group at risk of functional decline [2]. Such an instrument could be used as a screening tool in clinical practice and research to get an approximate value of VA in addition to other measurements of frailty. It is also important to get an approximate value of those who have moderate or severe visual impairment to be able to estimate visual impairment when there is no immediate access to an optometrist/ophthalmologist or in the elderly person's proximity. The purpose of such an instrument is to get an approximate value that can indicate visual impairment in terms of visual acuity. The measure could be used as a guide for further planning for geriatric assessments in care and rehabilitation. It could also be used to show the need for referring to further eye examinations and low vision rehabilitation. The aim of this study was to evaluate the construct validity and describe the sensitivity, specificity and predictive value of two short charts of visual acuity in order to find an instrument that can identify those with a VA level of ≤0.5 and that can approximately distinguish between early, moderate and severe signs of visual impairment.

Methods

Study design

The level of inter-instrument agreement between scales refers to construct validity [27] and was evaluated from an ophthalmologist's assessments of visual acuity, which forms the gold standard, and assessments from two short charts of visual acuity, made by a research assistant, to identify the agreement between different ways of measuring visual acuity. The study was blinded, since the information about the ophthalmologist's assessment of VA and the participant's eye disease was collected from the ophthalmologist after the research assistant's assessments. To identify those with visual acuity ≤0.5, the sensitivity, specificity and predictive value were calculated on the basis of the ophthalmologist's assessments of visual acuity and the assessments from the two short charts of visual acuity.

Participants

The participants were recruited at an open eye clinic at a specialist hospital in the southwestern part of Sweden. The inclusion criteria were 65 years and older and some kind of diagnosed, age related eye disease. The participants had to be living in their own homes and had to understand and make themselves understood in Swedish. A total of 48 persons were asked and accepted participation in the study, but five persons dropped out, n = 43. Seventy-seven percent of the participants had more than one eye disease, 74% were women, and the majority, 70%, had no social services. Three quarters of the participants described their self-reported general global health [28] as good or very good (Table I). The age of the persons who dropped out ranged between 73 and 91 years, median 84 years, and two of the five dropouts were men. The characteristics of the dropouts did not differ from those of the 43 participants. The main causes of dropouts were sickness, too busy or travel.

Instruments of visual acuity

The KM chart is a letter chart based on Monoyer standard that is composed of seven letters from the Roman alphabet [29] and it has been established that these letters have similar legibility [30]. The letter size used is four units wide and five units high and the inter-letter space is equal to one letter. The KM chart is arranged in rows with an arithmetical progression of VA recordings from 0.1 to 1.0. The eyes are tested separately and binocularly, and VA is assessed from <0.1 to 1.0 in ten scale steps. The VA level was recorded as the last row for which seven of ten letters were read correctly [29], which corresponds to clinical practice. Two different KM charts were used in this study: 5 m KM chart (gold standard) and a 1 m KM chart. The KM chart can be manufactured for different measurement distances and the chart adjusted for 1 m is 21 × 29.5 centimetres.

The Visual Acuity Screening Test is made up of a single optotype, a stylized letter E, in various orientations (Tumbling E optotypes) [31]. The individual's task is to indicate where the open side of the symbol is located – top, bottom, left or right – from the individual's point of view. A set is successfully completed when the open side of every symbol in the test has been correctly identified. The test ends when two mistakes are made in two consecutive sets, and the individual's score is the value assigned to the last set in which he/she managed to identify every symbol correctly. The eyes are tested separately

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Table I. Characteristics of the study participants.

	Participants (n = 43)
Age (years)	
Md	82
Range	67–94
Gender	
Men	11 (26%)
Women	32 (74%)
Eye disease[a]	
Glaucoma	27 (63%)
Cataract	32 (74%)
AMD	13 (30%)
Others	20 (47%)
Marital status	
Living with partner/married	20 (46%)
Living alone	23 (54%)
Living arrangements	
House	10 (23%)
Flat	33 (77%)
Social service	
Yes	13 (30%)
No	30 (70%)
Self-reported health	
Very good	4 (9%)
Good	28 (65%)
Poor	10 (23%)
Very poor	1 (2%)

[a]All participants can be diagnosed as having more than one eye disease.

and binocularly [32], and VA is assessed from 0.02 to 1.0 in 15 scale steps. In the present study, VA levels 0.02–0.08 are included in level <0.1, meaning that the scale in this study has 10 scale steps. A Visual Acuity Screening Test chart for 1 m distance was used in this study. The Visual Acuity Screening Test has been applied in a European project focusing on healthy aging [33].

Procedure

Persons at a routine visit were asked by an ophthalmologist to participate in the study. The participants were selected consecutively according to the inclusion criteria. The ophthalmologist tested the participant's VA with a KM chart adjusted for 5 metre distance with best refraction and the participant's distance glasses were thereafter corrected if needed.

After the visit to the ophthalmologist, a research assistant contacted the participants by telephone and additional information was given about the study. Informed consent was obtained after both verbal and written information was given. The persons (n = 43) were assessed in their own homes approximately 2 weeks after the ophthalmologist visit. The home visits were made during daytime and lasted between 1 and 1.5 h. The research assistant first interviewed the participants according to a structured protocol that consisted of questions about marital status, living arrangements, social services and self-reported global health [28]. No person was excluded on the grounds of a perceived change in vision since the ophthalmologist measurement.

The research assistant then assessed illuminance and luminance using a photometer (Hagner model S2, Hagner AB, Solna, Sweden). This was done to ensure finding the brightest area for measuring VA so that the assessments could be made under optimal lighting conditions given the circumstances. The participant's VA was assessed with the 1 m KM chart and the Visual Acuity Screening Test, and these assessments lasted between five and ten minutes. Participants that had distance eyeglasses were requested to use them. The research assistant had received training in assessing VA with the 1 m KM chart and the Visual Acuity Screening Test by the ophthalmologist. The entire data collection was done over a 3-month period.

Data analysis

The level of inter-instrument agreement (construct validity) between the assessments made with the 1 m KM chart, the Visual Acuity Screening Test and the 5 m KM chart (gold standard) was evaluated focusing on the level of order consistency between the assessments made by the two raters, the ophthalmologist and the research assistant. A high level of order consistency is a prerequisite for the inter-changeability of scale. The level of order consistency is defined by the coefficient of monotonic agreement, MA. MA is the difference between the proportions of ordered and disordered pairs of observations and can be calculated by MA = 1-2D. D is a measure of disorder. The MA can range between −1 and 1. The maximum value, MA = 1, is obtained when there is a total order consistency of the observations [27,34]. This means that values close to 1 have the highest agreement, and there is complete agreement between pairs of observations when MA = 1. Percentage agreement (PA) was used to express the proportion of the inter-instrument agreement between the paired assessments. RP (Relative Position) and RC (Relative Concentration) were measured to complement the measurement of inter-instrument agreement between the scales. The possible value of RP and RC range from −1 to 1, and zero values mean a lack of agreement/systematic disagreement [27,34].

The assessments with the two KM charts and the Visual Acuity Screening Test were made on a ten-point scale which was reduced to a four-point scale. The definition of visual acuity levels in this four-point scale was made as follows: VA levels *moderate visual impairment* (<0.3–0.1) and *severe visual impairment* (<0.1) were used according to WHO's definition of visual acuity for blindness and low vision [24,25]; WHO's VA level 1.0–0.3, *mild and no visual impairment*, was divided into two VA levels, one level (1.0–0.6) had the same label (*mild and no visual impairment* 1.0–0.6) while the other VA level 0.5–0.3 was labelled *early signs of visual impairment*. This was done in order to identify levels of acuity that risk compromising the performance of daily activities among older people [12,23,26]. These four cut-off VA levels were used in the calculations of inter-instrument agreement between the scales. The VA of the best eye with the best correction was the value chosen as the VA parameter in the ophthalmologist assessment, and the VA of the best eye using the participant's own distance glasses was the value chosen as the VA parameter at the home visit.

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The sensitivity, specificity and predictive values were calculated to describe the probability of the instrument [35] to screen out those with a VA of ≤0.5. Sensitivity means the proportion of assessments that correctly identified VA ≤ 0.5 by the 1 m KM chart or the Visual Acuity Screening Test. Specificity means the proportion of assessments that correctly identified VA 0.6–1.0 by the 1 m KM chart or Visual Acuity Screening Test. The positive and negative predictive values give an assessment of the clinical usefulness of the test [35]. The positive predictive value is the proportion of persons correctly assessed as having visual impairment by the 1 m KM chart or the Visual Acuity Screening Test. The negative predictive value is the proportion of persons correctly assessed as not having visual impairment by the 1 m KM chart or the Visual Acuity Screening Test.

Results

The level of order consistency (MA) was 0.964 for the four-point scale of the 5 m KM chart and the 1 m KM chart (Figure 1a) and 0.948 for the 5 m KM chart and the Visual Acuity Screening Test (Figure 2a). There was an inter-instrument agreement of 44% (PA) for the 5 m KM chart and the 1 m KM chart (Figure 1a) and of 37% (PA) for the 5 m KM chart and the Visual Acuity Screening Test (Figure 2a). The results of these inter-scale comparisons showed a high level of order consistency (high MA) and a low PA, which indicates a systematic disagreement between the two four-point sets of data (Figure 1a and 2a). The ophthalmologist systematically assessed the individuals as having higher VA levels than the research assistant. This is confirmed by the RP value of −0.32 for the 5 m KM chart and the 1 m KM chart and −0.36 for the 5 m KM chart and the Visual Acuity screening test. The systematic disagreement in position between the raters is illustrated by the convex shape of the ROC curves for the 5 m KM chart and the 1 m KM chart (Figure 1b) and for the 5 m KM chart and the Visual Acuity Screening Test (Figure 2b). The RC value was 0.04 for the 5 m KM chart and the 1 m KM chart. The RC value was 0.14 for the 5 m KM chart and the Visual Acuity Screening Test and showed a slight systematic disagreement in concentration between the two scales.

The sensitivity, which is the proportion of assessments made with the 1 m KM chart or the Visual Acuity screening test that correctly identified visual impairment (VA ≤ 0.5), was 100%. The specificity, that is, the proportion of assessments that correctly identified normal vision (VA > 0.6–1.0) was 38% for the 1 m KM chart and 25% for the Visual Acuity Screening Test. The positive predictive value, i.e. the proportion of persons with visual impairment (VA ≤ 0.5) who were identified as having visual impairment, was 73% for the 1 m KM chart and 69% for the Visual Acuity Screening Test. The negative predictive value, which is the proportion of persons with normal vision (VA > 0.6–1.0) who were identified as having normal vision, was 100% for both instruments (Table II).

Discussion

Both short charts of visual acuity, the 1 m KM chart and the Visual Acuity Screening Test, used for assessment of visual function in the home environment could identify the individuals with visual impairment. The results showed that we can expect the two short charts of visual acuity to distinguish 100% of those with a visual impairment (VA ≤ 0.5). The instruments have good positive and negative predictive values for the 1 m KM chart (73% and 100%) and for the Visual Acuity Screening Test (69% and 100%) which is very good [35]. This is important since research [9,12,13,23,26] shows that older people with visual impairment already tend to have problems in daily activities when they have early signs of visual impairment (VA ≤ 0.5). Identifying these is thus important so that rehabilitation can start early. Research shows that the earlier the intervention starts, and the less serious the activity limitations are, the better are the conditions for the ability to retain activity [3,17,18,23,36].

The level of ordering (MA) was high, which means that there was no change in ordering of the participant's level of visual acuity between the short charts of visual acuity and gold standard. This is an important prerequisite for the construct validity (inter-instrument agreement) of the scales [27,34]. However, the results show a low PA between the 5 m KM chart and the 1 m KM chart or the Visual Acuity Screening Test, which indicates a systematic disagreement

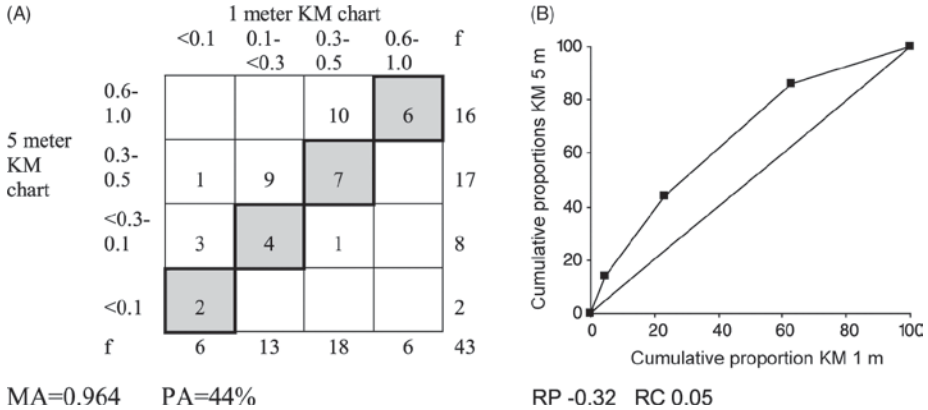


Figure 1. (a) Joint distribution of paired assessments of visual impairment measured by 1 m KM chart and 5 m KM chart. f, frequencies. (b) The ROC curve for the systematic change in visual impairment between the 1 m KM chart and 5 m KM chart (n = 43). RP, relative position, RC, relative concentration.

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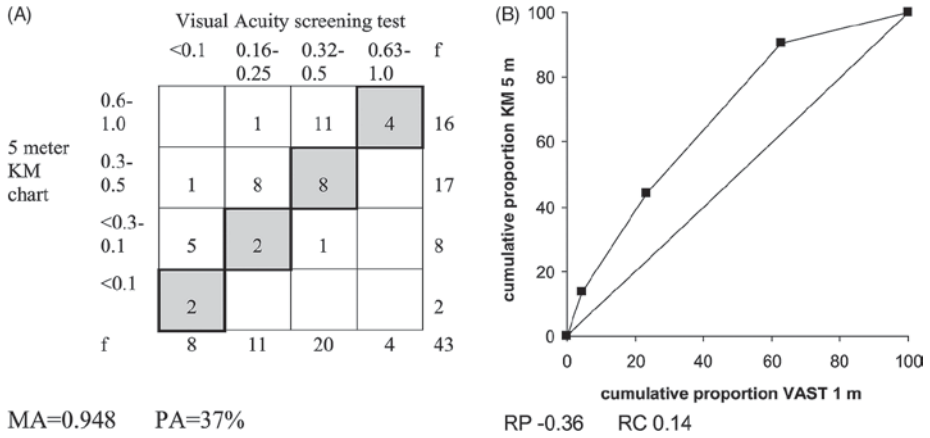


Figure 2. (a) Joint distribution of paired assessments of visual impairment measured by Visual Acuity Screening Test (VAST) and 5 m KM chart. f, frequencies. (b) The ROC curve for the systematic change in visual impairment between Visual Acuity Screening Test and 5 m KM chart (n = 43). RP, relative position, RC, relative concentration.

Table II. Sensitivity, specificity and predictive values (percent) for 1 m KM chart and Visual Acuity Screening Test.

Values	1 m KM chart	Visual Acuity Screening Test
Sensitivity	100	100
Specificity	38	25
Positive predictive value	73	69
Negative predictive value	100	100

[27,34] between the four-point sets of data. This pattern of systematic disagreement is explained by the fact that the ophthalmologist assessed the persons to have higher VA levels than in the assessment by the research assistant, which was confirmed by the RP values [34]. It can be argued that this difference is due to the discrepancy in test distance between the instruments and that the charts are different. Another reason can be that the ophthalmologist and research assistant have different knowledge and experience of assessing VA levels. A more potent explanation is that the ophthalmologist's assessments were made with more sophisticated assessment instruments and under optimal lighting conditions, while the assessments made by the research assistant were made in home environments. Even though the brightest area for measuring VA was chosen, lighting conditions are most likely poorer in home environments than in the ophthalmologist's office. Yet another explanation for the systematic disagreement can be that the ophthalmologist made the assessments with the best refraction and, even if the participants' were prescribed to buy new distance glasses if needed, we do not know if this was done. To summarize, under these circumstances, there should be a difference between the assessments but, since the intention is to approximately distinguish those with visual impairment, this disagreement does not seriously hamper the reliability of the instruments but should be noticed when used.

One problem with the assessments was that persons who have no visual impairment could be assessed as having impaired vision. The short charts identified fewer participants having normal vision compared to the ophthalmological assessments. This means that a visual screening test in an everyday environment might overestimate visual impairment slightly. Considering the importance of identifying persons at risk of visual frailty, an underestimation could be potentially more hazardous. Measures for sensitivity and specificity are not sufficient to describe the probability of an instrument; estimates of predictive values to correctly identify those with visual impairments are also needed [35]. The instruments have good positive and negative predictive values for the 1 m KM chart (73% and 100%) and for the Visual Acuity Screening Test (69% and 100%). The instruments are therefore useful tools when there is a need of screening for visual impairment when there is no access to an ophthalmologist or optometrist. However, it is important to consider that these short charts of visual acuity cannot replace a complete eye examination or complete ophthalmological examination.

Although visual impairment has been shown to be a marker of frailty in the older population, it is seldom included as an assessment for older people in geriatric care or geriatric rehabilitation [20]. These short charts can be used to identify older persons that have visual impairment as one part of the CGA and is one indicator that should be measured when assessing frailty [20]. They should only be seen as guides for further planning and form the basis for CGA in care and rehabilitation or be used to show the need for referral to eye examination.

The short charts of visual acuity differ in some areas but are mostly similar. The design of the instruments differs, the 1 m KM chart is constructed by Roman alphabet letters and the Visual Acuity Screening Test is constructed by tumbling E optotype. An advantage for the Visual Acuity Screening Test, with tumbling E optotype, is that it can be used to test persons with illiteracy and populations not familiar with the Roman alphabet [32]. The similarities are that the amount of time necessary for the assessment was experienced as the same, the instruments seem equally easy to understand and the instructions before the assessments were alike. Both short charts

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have a small format, which makes them easy to transport. In addition, both are standardized for assessments at a distance of 1 m and can thus be useable in environments other than the eye clinic for instance home environments. Assessments in home environments can be of great advantage for those frail elderly persons with mobility problems.

A limitation of both short charts of visual acuity as markers of frailty in the aging population is that they measure only one parameter in visual impairment, visual acuity. However, visual field defect, another parameter in visual impairment, is not included in this screening. Eye diseases in this age group that are characterized by visual field defects such as glaucoma where decline in visual acuity comes at a later stage of the disease will not be found in an early stage by using these short charts of visual acuity [37]. The argument behind the need of a short chart of visual acuity as markers of frailty was to identify persons that are at risk of developing dependency in daily activities, and persons with glaucoma do not report problems in daily life until at a later stage [37]. Visual acuity has shown to be strongly correlated to problems in daily activities compared to other visual impairment parameters [37]. Thus, recognizing the limitation mentioned above, these short charts of visual acuity should be valuable tools as markers of frailty.

In conclusion, these short charts of visual acuity have good construct validity (inter-instrument agreement) considering that it was held in the persons' homes and a good predictive value for identifying those who have a VA level of ≤0.5. The inter-instrument agreement between the 5 m KM chart and the 1 m KM chart or the Visual Acuity Screening Test showed that the 1 m KM chart has better values in the MA, PA, RP and RC. This means that the 1 m KM chart has the best agreement with the 5 m KM chart, which was used in this study as the gold standard.

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